

Fractals

Infinite Details from Simple Rules

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CSE 200 Presentation

History

So, You're Designing All the Trees...

Imagine you're a game developer. You've just been handed one job: design **every tree** in the game — all of them.

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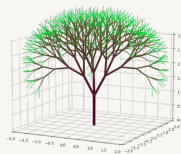
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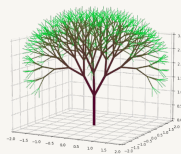
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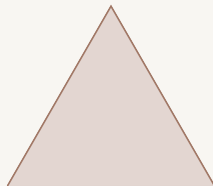


Or, you write **one rule** — “a branch splits into two smaller branches” — and let that rule call itself, again and again, at a smaller scale every time.

Trunk becomes branch, branch becomes twig, twig becomes leaf — **same rule, every time**. The engine genuinely can't tell the whole tree apart from one tiny piece of it.

Déjà Vu?

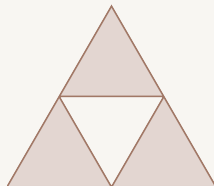
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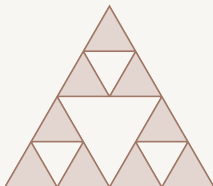


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Here's the simplest possible version of that same trick: draw a triangle.

Cut a smaller triangle out of the middle.

Now do that again, inside each of the three triangles you've got left.



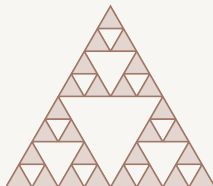
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Weird, right? That's a **fractal** — a shape built out of smaller copies of itself.

What are Fractals?

Definition

A **fractal** is a geometric shape containing detailed structure at arbitrarily small scales, usually having a **fractal dimension** strictly exceeding the **topological dimension**.

Key Characteristics:

- **Scale invariance**: Details at every level of magnification
- **Fractal dimension**: More than the *conventional* dimension
- **Infinite complexity**: Generated by simple, recursive rules
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*"Clouds are not spheres, mountains are not
cones, coastlines are not circles..."*

— **Benoit Mandelbrot**

Mandelbrot's Question

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Mandelbrot's insight: this isn't a flaw in the coastline — it's a sign that classical length and area are **the wrong tools** for something like this. What a coastline actually needed was exactly the kind of **“useless” geometry** mathematicians had quietly been building for a century already — without knowing why.

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For nearly a century, these were dismissed as **beautiful but useless**. Mandelbrot was the one who realized they weren't useless at all — he gave the whole family a name: **fractal**.

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And that triangle? The exact one you watched me draw a few minutes ago.

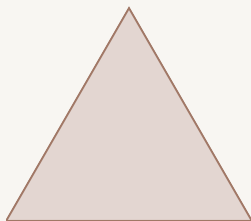
Some Fractals

The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

Construction Process

1. Start with an equilateral triangle



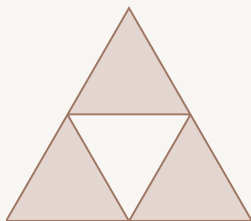
Iteration 0

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Let's construct the Sierpinski Triangle:

Construction Process

1. Start with an equilateral triangle
2. Remove the central triangle (triangle connecting midpoints of the sides)



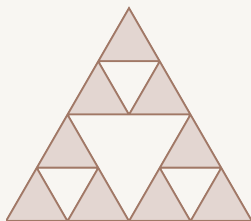
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Let's construct the Sierpinski Triangle:

Construction Process

1. Start with an equilateral triangle
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3. Repeat for each triangle



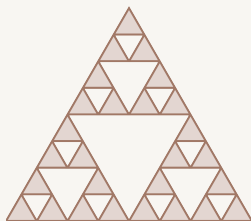
Iteration 2

The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

Construction Process

1. Start with an equilateral triangle
2. Remove the central triangle (triangle connecting midpoints of the sides)
3. Repeat for each triangle
4. After infinite iterations, we get the Sierpinski Triangle



Iteration 3

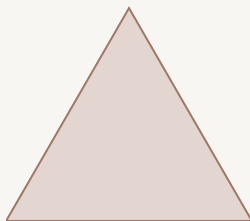
Area & Perimeter

The Sierpinski Triangle: Area

Now let's find the area of triangle(s):

Area Calculation

$$A_0 = \frac{1}{2} \cdot \sin(60^\circ) \cdot a^2 = \frac{\sqrt{3}}{4} a^2$$



Iteration 0

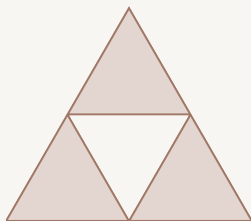
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$$\begin{aligned} A_1 &= A_0 - \frac{1}{2} \cdot \sin(60^\circ) \cdot \left(\frac{a}{2}\right)^2 \\ &= A_0 - \frac{\sqrt{3}}{16} a^2 = A_0 - \frac{1}{4} A_0 \\ &= \frac{3}{4} A_0 \end{aligned}$$



Iteration 1

We cut out one fourth of the area at each step.
So, naturally area is reduced by $\frac{3}{4}$ at each step.

The Sierpinski Triangle: Area

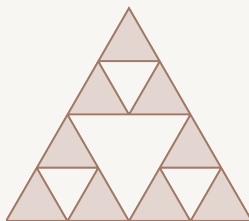
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Area Calculation

$$A_0 = \frac{1}{2} \cdot \sin(60^\circ) \cdot a^2 = \frac{\sqrt{3}}{4} a^2$$

$$A_1 = \left(1 - \frac{1}{4}\right) A_0 = \frac{3}{4} A_0$$

$$A_2 = \left(1 - \frac{1}{4}\right) A_1 = \left(\frac{3}{4}\right)^2 A_0$$



Iteration 2

We cut out one fourth of the area at each step.
So, naturally area is reduced by $\frac{3}{4}$ at each step.

The Sierpinski Triangle: Area

Now let's find the area of triangle(s):

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$$A_0 = \frac{1}{2} \cdot \sin(60^\circ) \cdot a^2 = \frac{\sqrt{3}}{4} a^2$$

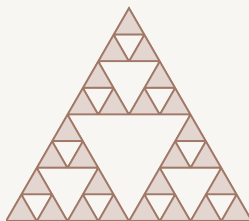
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$$A_n = \left(1 - \frac{1}{4}\right)^n A_0 = \left(\frac{3}{4}\right)^n A_0$$

$$\lim_{n \rightarrow \infty} A_n = 0$$



Iteration 3

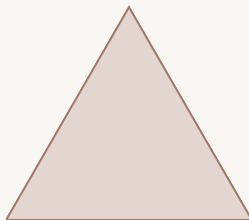
The area of the Sierpinski Triangle is zero even though it is a 2D shape.

The Sierpinski Triangle: Perimeter

Now let's find the perimeter:

Perimeter Calculation

$$P_0 = 3a$$



Iteration 0

The Sierpinski Triangle: Perimeter

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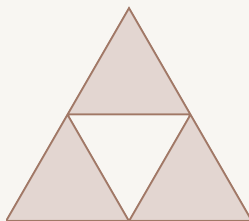
Perimeter Calculation

$$P_0 = 3a$$

$$P_1 = P_0 + 3 \cdot \frac{a}{2} = \frac{9}{2}a = \frac{3}{2}P_0$$

We add half of the previous perimeter at each step. This is because we add the same number of triangles as the last step, but each triangle is half the size of the previous one.

As a result, the perimeter grows by a factor of $\frac{3}{2}$ at each step.



Iteration 1

The Sierpinski Triangle: Perimeter

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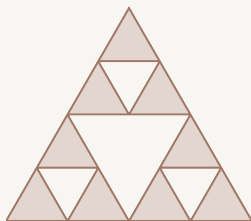
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Iteration 2

The Sierpinski Triangle: Perimeter

Now let's find the perimeter:

Perimeter Calculation

$$P_0 = 3a$$

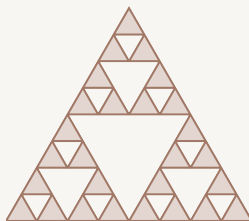
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...

$$P_n = \frac{3}{2}P_{n-1} = \left(\frac{3}{2}\right)^n P_0$$

$$\lim_{n \rightarrow \infty} P_n = \infty$$



Iteration 3

The Sierpinski Triangle has infinite perimeter but zero area.

We've now seen something strange, and precise:

- A **2D shape** (Sierpinski Triangle) with **zero area**.
- A shape built from a rule so simple a child could apply it by hand.
- No matter how far we zoom in, there's always more detail.

Handoff to Speaker 2

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If it isn't a line, and it isn't a surface — what **is** its true dimension?

That number is exactly what Hausdorff was reaching for in 1918 — and it's where Speaker 2 picks up the story.

Fractal Dimension

Let's talk about a property of fractals...



Fractal Dimension

What does it mean for a shape to have a **non-integer** dimension?

Dimensions We Know

We are all familiar with integer dimensions:



Dimensions We Know

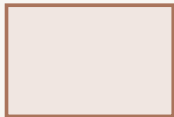
We are all familiar with integer dimensions:



Line



1D



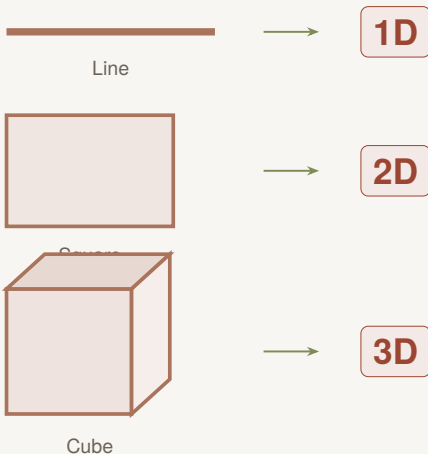
Square



2D

Dimensions We Know

We are all familiar with integer dimensions:



Dimensions are always whole numbers... right?

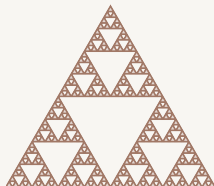
What About This One?

What dimension does this
shape have?

Any guesses?

- It lives in 2D space...
- But it has **zero area**
- And **infinite perimeter**

It's not quite 2D, but it's clearly more than
1D...



Sierpinski Triangle

What About This One?

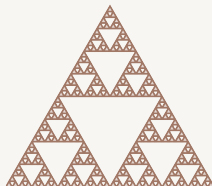
Sierpinski Dimension

The Sierpinski Triangle has a fractal dimension of:

$$D \approx \mathbf{1.585}$$

!!

A **non-integer** dimension — between a line (1D) and a filled surface (2D).



$$D = \log 3 / \log 2 \approx 1.585$$

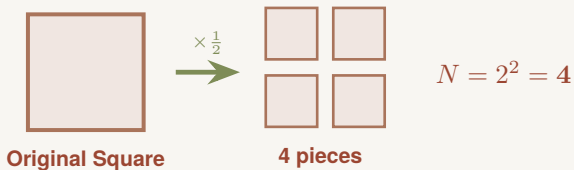
Self-Similarity & Scaling

Scaling a Line ($D = 1$): Scale by $r = \frac{1}{2} \implies$ **2 pieces.** Scale again by $\frac{1}{2} \implies 2 \times 2 =$ **4 pieces.**



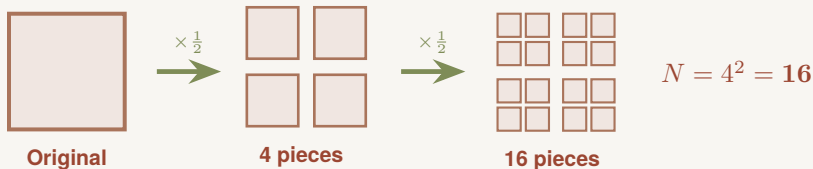
Self-Similarity & Scaling

Scaling a Square ($D = 2$) — Step 1: Scale by $r = \frac{1}{2} \implies$ **4 self-similar pieces** (2×2 grid).



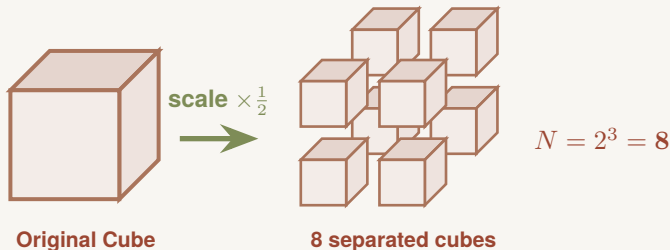
Self-Similarity & Scaling

Scaling a Square ($D = 2$) — Step 2: Scale each piece by $\frac{1}{2}$ again $\implies 4 \times 4 = 16$ self-similar pieces (4×4 grid)!



Self-Similarity & Scaling

Scaling a Cube ($D = 3$): Scale by $r = \frac{1}{2} \implies$
8 self-similar pieces ($2 \times 2 \times 2$ grid in 3D).



Self-Similarity & Scaling

Do You See a Pattern?

Each scaling step multiplies pieces by $(1/r)^D$!

Shape	Scale	1st Step	2nd Step	D
Line	$1/2$	$2 = 2^1$	$4 = 2^2$	1
Square	$1/2$	$4 = 2^2$	$16 = 2^4$	2
Cube	$1/2$	$8 = 2^3$	$(64 = 2^6)$	3



$$D = ?$$

What if the shape is a **fractal**?

The Pattern

$$\text{Number of pieces} = \left(\frac{1}{r}\right)^D$$

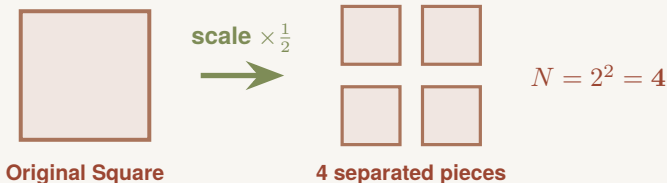
The Self-Similarity Rule

After scaling by factor r , the number of self-similar pieces is:

$$N = \left(\frac{1}{r} \right)^D$$

Square ($D = 2$) with scale factor $r = \frac{1}{2}$:

$$N = \left(\frac{1}{1/2} \right)^2 = 2^2 = 4 \text{ pieces}$$



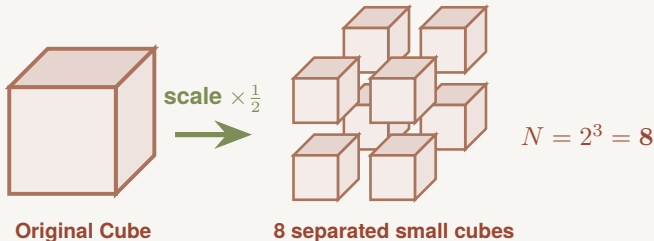
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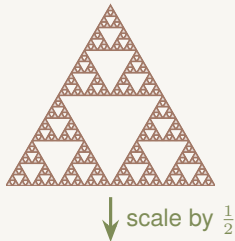
Can we use this same rule to find the dimension of a **fractal**?

Dimension of the Sierpinski Triangle

Let's apply the self-similarity rule to the **Sierpinski Triangle**.

Step 1

Scale the Sierpinski Triangle by $\frac{1}{2}$.

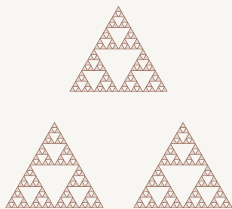


Dimension of the Sierpinski Triangle

Step 2 — Self-Similar Pieces

We get exactly **3** self-similar copies of the original triangle.

Each smaller triangle is a perfect replica, scaled by $r = \frac{1}{2}$.



3 copies

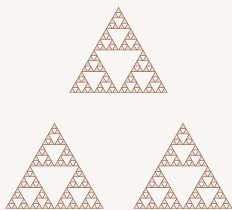
Dimension of the Sierpinski Triangle

Step 3 — The Equation

Using $N = \left(\frac{1}{r}\right)^D$:

$$3 = \left(\frac{1}{1/2}\right)^D$$

$$3 = 2^D$$



3 copies

Dimension of the Sierpinski Triangle

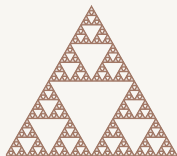
Step 4 — Solving for D

$$2^D = 3$$

$$D \log 2 = \log 3$$

$$D = \frac{\log 3}{\log 2}$$

$$D \approx \mathbf{1.585}$$



$$D \approx 1.585$$

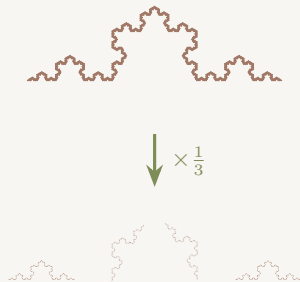
Not an integer! — strictly between 1D (line) and 2D (surface).

Similarly: The Koch Curve

The **Koch Curve** can be divided into **4** self-similar pieces, each scaled by $r = \frac{1}{3}$.

Setup

- Scale factor: $r = \frac{1}{3}$
- Self-similar pieces: $N = 4$



4 copies

Similarly: The Koch Curve

Koch Dimension

$$N = \left(\frac{1}{r}\right)^D$$

$$4 = 3^D$$

$$D = \frac{\log 4}{\log 3}$$

$$D \approx \mathbf{1.262}$$



$$D \approx 1.262$$

Also **non-integer** — between a line (1D) and a surface (2D), but closer to a line than the Sierpinski Triangle.

The General Theory: Box-Counting Dimension

The **Self-Similarity Dimension** works beautifully for self-similar fractals. But what about more complex shapes?

Box-Counting Dimension

The fractal dimension D is defined as:

$$D = \lim_{\varepsilon \rightarrow 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}$$

where $N(\varepsilon)$ is the number of boxes of side length ε needed to cover the fractal.

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where $N(\varepsilon)$ is the number of boxes of side length ε needed to cover the fractal.

Intuition

- Cover the fractal with a grid of boxes of size ε
- Count how many boxes $N(\varepsilon)$ the fractal touches
- Shrink $\varepsilon \rightarrow 0$ and observe how $N(\varepsilon)$ grows
- The **rate of growth** tells us the dimension

For a deeper dive, see [3Blue1Brown: Fractals are typically not self-similar](#)

Handoff to Speaker 3

We've now uncovered a remarkable property of fractals:

- They have **non-integer dimensions** — genuinely “between” the familiar 1D, 2D, 3D.
- The Sierpinski Triangle lives in dimension $D \approx 1.585$.
- The Koch Curve lives in dimension $D \approx 1.262$.
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That's exactly where Speaker 3 takes us — fractals in nature, art, and computation.

Fractals in the Real World

We know what a fractal is, and how to measure it...



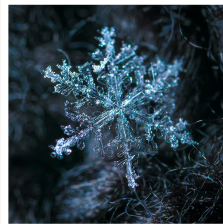
Fractals in the Real World

Where does all of this actually **show up**?

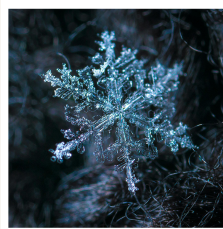
Nature draws with one rule, repeated



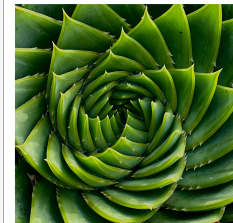
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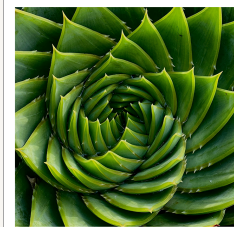
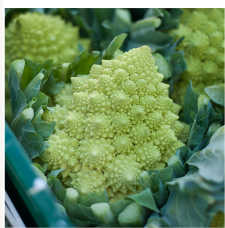
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Broccoli, ice, leaves, cones — the **same shape at every zoom**.

A tree built from nothing but squares

3 levels



Pythagorean tree, after A. E. Bosman (1942), drawn at 45° .

A tree built from nothing but squares

3 levels



7 levels



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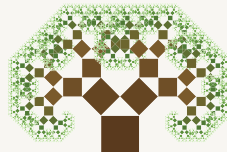
3 levels



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12 levels



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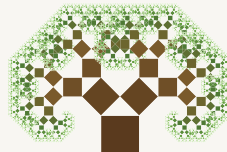
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Stand a **right triangle** on each square's top edge and grow a new square on each leg — so $a^2 + b^2 = c^2$ sits at every single joint.

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Stand a **right triangle** on each square's top edge and grow a new square on each leg — so $a^2 + b^2 = c^2$ sits at every single joint.

Nobody designed a single branch. **One rule**, repeated twelve times, and a tree appears.

Pythagorean tree, after A. E. Bosman (1942), drawn at 45° .

tilted rule



tilted rule



- At 45° the two branches match **exactly** — too perfect to be real

tilted rule



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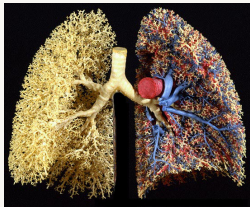
tilted rule



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One number changed — and a diagram turns into a plant.

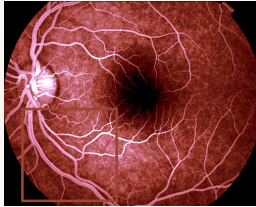
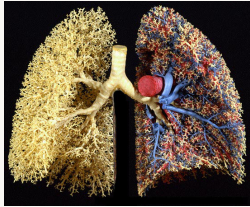
Branching packs a huge surface into a small space



airways and blood vessels of the lung vessels in the retina the same vessels, zoomed in

Lung branching figures: Weibel (1963).

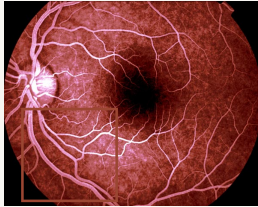
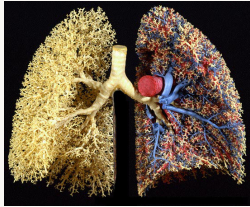
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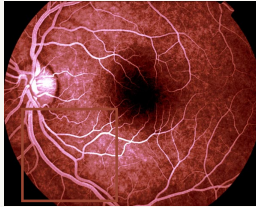
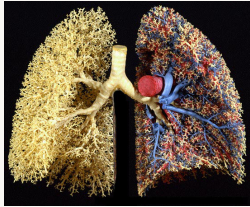
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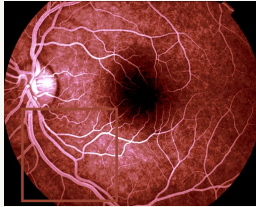
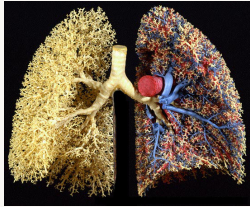


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- Lungs: about **23 splits** $\Rightarrow \sim 70 \text{ m}^2$ of surface packed inside ~ 5 litres

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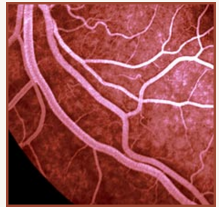
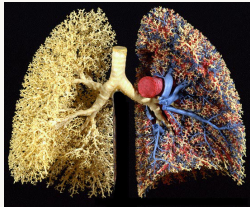


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Branching again and again is nature's cheapest way to get **area out of volume**.

Lung branching figures: Weibel (1963).

The whole fern is just four rules

Every dot is placed by **one of four rules**, chosen at random. Colour each dot by the rule that drew it:



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Rule 2 draws a **small copy of the whole plant**. That single line is where all the detail comes from.

Fractal compression: a whole plant in 28 numbers

2,000 dots



Barnsley fern: an iterated function system (Barnsley, *Fractals Everywhere*, 1988), plotted by the chaos game.

Fractal compression: a whole plant in 28 numbers

25,000 dots



- Four rules, each a **shrink-and-tilt** of the whole picture, applied at random

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Store the **rule**, not the pixels — the idea behind **fractal image compression**.

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Game worlds are grown from a rule, not stored

1 layer of bumps



Terrain generated here by summing 7 layers of noise, each half the size of the last (fractal Brownian motion).

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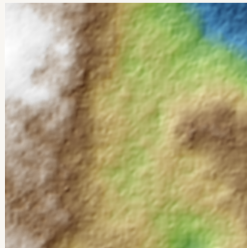
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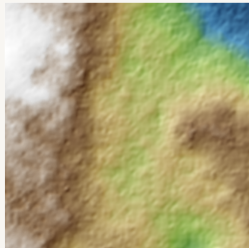
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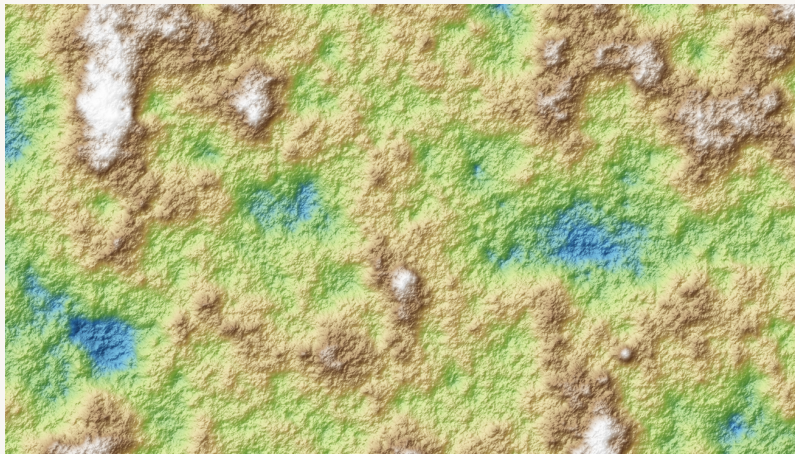
7 layers



Add **smaller bumps at every scale** — the Koch idea — and a handful of numbers becomes an endless mountain range. Store the rule, not the map.

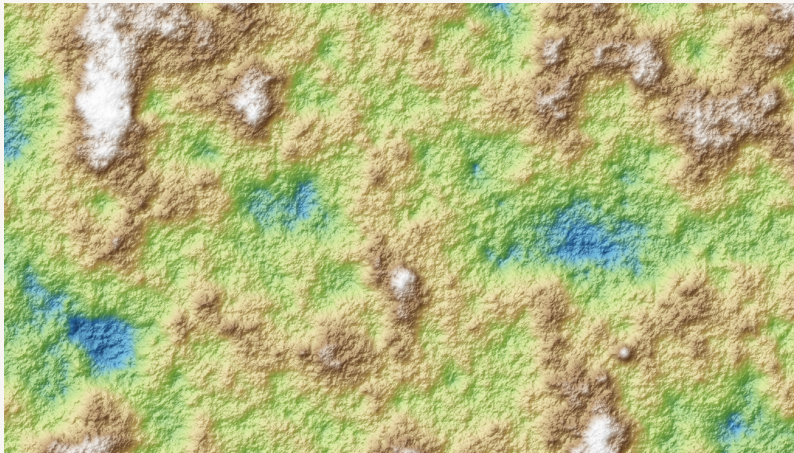
Terrain generated here by summing 7 layers of noise, each half the size of the last (fractal Brownian motion).

Keep stacking layers and you get a world



15 layers of value noise, each half the width and $0.55 \times$ the height of the one before.

Keep stacking layers and you get a world



Same rule, **15 layers** instead of 7 — ridges, valleys, snow lines and lakes. Not one of them was drawn by hand.

15 layers of value noise, each half the width and $0.55 \times$ the height of the one before.

One honest catch: real fractals stop

- A fern looks like a fern ... until you reach a single **cell**

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A fractal is a **model** of nature — and every good model knows the range where it works.

One rule, repeated, explains what smooth geometry couldn't

Part 1 — the surprise



zero area,
infinite edge

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infinite edge

Part 2 — the number

$$D = \frac{\log N}{\log(1/r)}$$

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tilted rule



trees, lungs, ferns,
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*“Clouds are not spheres, mountains are not cones,
coastlines are not circles...” — Mandelbrot*

Roughness has a number, and simple rules make it.

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Thank you — questions welcome.

- B. Mandelbrot, “How long is the coast of Britain? Statistical self-similarity and fractional dimension,” *Science*, 1967.
- B. Mandelbrot, *The Fractal Geometry of Nature*, 1982.
- E. R. Weibel, *Morphometry of the Human Lung*, Springer, 1963.
- M. F. Barnsley, *Fractals Everywhere*, Academic Press, 1988 (iterated function systems, the fern).
- A. E. Bosman, *Het wondere onderzoekingsveld van de vlakke meetkunde*, 1957 (Pythagorean tree, drawn 1942).
- A. Fournier, D. Fussell, L. Carpenter, “Computer rendering of stochastic models,” *Comm. ACM*, 1982 (fractal terrain).
- 3Blue1Brown, “Fractals are typically not self-similar” (video).